Hierarchical Spatial Synergy Network for Landslide Detection in Satellite Remote Sensing Imagery

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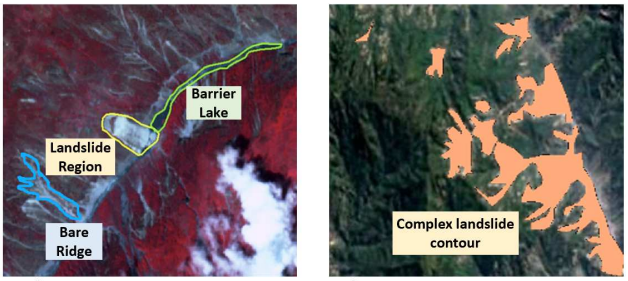
***Abstract*—Landslide detection is essential in natural disaster monitoring. However, in satellite remote sensing imagery, landslide areas exhibit complex spatial patterns and are subject to noise interference from bright backgrounds. It presents significant challenges for accurate landslide detection. To address these issues, the Hierarchical Spatial Synergy Network (HSSNet) is proposed. HSSNet designs a multiscale synergy attention module to enhance discriminative features of the landslide areas by optimizing multiscale contextual features while suppressing noise from bright backgrounds. Additionally, HSSNet introduces a deformable spatial refining module that adaptively extracts irregular spatial structure features, improving the representation of the complex spatial structure of landslide regions. Experimental results demonstrate that HSSNet achieves comparable performance to mainstream methods.**

Keywords- Landslide detection, remote sensing image, semantic segmentation.

I. INTRODUCTION

Landslides are geological disasters resulting from the downward movement of rock or soil along slip surfaces due to gravity [1]. They pose severe threats to power transmission infrastructure and directly endanger human lives and property. Anhui Province, located in the transitional zone between the Jianghuai Hills, Dabie Mountains, and the southern Anhui ranges, is highly prone to such events due to its complex terrain and monsoon climate. In 2024 alone, 1,154 landslides were reported, causing economic losses exceeding 1.7 billion RMB[1]. Therefore, timely and accurate landslide detection is essential in disaster prevention and mitigation.

Landslide detection focuses on pixel wise classification to ascertain if a pixel is within a landslide region. Early methods primarily relied on meteorological forecasting and field investigations by geologists, identifying landslide risks by observing surface cracks, abnormal vegetation, soil displacement, and other phenomena [2]. This approach was highly constrained by terrain and had high labor costs. As high-resolution satellite imagery technology evolves, satellites such as Sentinel-1/2 and Landsat-9 have provided vast amounts of multispectral remote sensing data, laying the data foundation for large-scale landslide monitoring [3]. However, efficiently extracting landslide features from massive remote sensing data remains a key bottleneck limiting the application of this technology.



a) The interference from high-intensity similar targets.

b) The landslide exhibits complex spatial structures.

Figure 1. The challenges of landslide detection.

With rapidly development in deep learning, some research defines the landslide detection as a classic semantic segmentation task. These algorithms leverage neural networks to learn discriminative features such the spectrum, texture, and deformation of landslide areas, significantly improving detection efficiency compared to traditional methods. Typically, landslide exhibits high reflectivity in remote sensing images, allowing these areas to be distinguished from features with lower reflectivity. However, as illustrated in Figure 1, the high reflectivity of bare land objects can cause them to be mistakenly identified as landslide areas, resulting in false positives. [6]. Furthermore, landslide regions typically exhibit irregular spatial structures. The regular feature sampling positions in standard convolutional neural networks often

[1] Notice of the Anhui Provincial Department of Natural Resources on Issuing the Geological Disaster Prevention Plan for Anhui Province in 2024, https://www.guangde.gov.cn/OpennessContent/show/3311643.html

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lead to blurred spatial activations, resulting in overly smooth boundary predictions [7].

To tackle these challenges, this paper develops the Hierarchical Spatial Synergy Network (HSSNet) for landslide detection in satellite remote sensing imagery. As shown in Figure 1, HSSNet adopts a symmetric encoder-decoder architecture to ensure consistent spatial feature learning. To mitigate the interference from high-intensity similar targets, HSSNet introduces the Multiscale Synergy Attention (MSA) module. MSA utilizes dilated convolution and spatial-channel attention mechanisms to optimize multi-scale features, boosting landslide detection while suppressing high-intensity background noise. HSSNet introduces the Deformable Spatial Refining (DSR) module to optimize the irregular spatial structures of landslide regions. Experiments on the CAS public dataset [8] show that HSSNet achieves comparable performance to mainstream methods.

The contributions are summarized below:

1. The Multiscale Synergy Attention is designed to identify landslide regions from complex backgrounds by integrating multi-scale dilated convolutions and global-local contextual features.
2. The Deformable Spatial Refining is proposed to optimize spatial activations and extract the irregular spatial structure of landslides by learnable positional offsets.
3. HSSNet demonstrates the performance on public CAS dataset.

II. Related Works

Geological hazards cause surface deformation and destruction, which alter the land's optical properties. These changes can be detected using satellite remote sensing imagery, making optical remote sensing a key tool for geological hazard assessment and disaster evaluation [1].

In recent years, Convolutional Neural Networks (CNNs) achieves significant success in landslide detection tasks due to their powerful feature learning capabilities[4 - 5]. Typically, Ding et al. [9] employed CNNs to extract features and use the Euclidean distance to determine landslide. Yu et al. [10] utilized CNNs for landslide detection and applied a region-growing algorithm to extract the landslide boundaries and center. Ghorbanzadeh et al. [11] explored CNNs parameter sensitivity on landslide detection. However, these studies primarily relied on relatively simple network architectures, which still face challenges in suppressing background interference and optimizing spatial structures.

To address these problems, Lei et al. [12] combined pyramid pooling to enhance small-scale landslide detection. Chen et al. [13] incorporated landslide susceptibility as prior knowledge for landslide detection. Liu et al. [14] proposed rapid landslide identification using residual U-Net. Lu et al. [15] leveraged attention U-Net further for LiDAR landslide recognition. Fang et al. [16] replaced the convolutional blocks with RegNet to improve landslide detection accuracy. Fang et al. [17] introduced a GAN-based Siamese architecture to generate landslide inventory maps without any preprocessing or post-processing steps. Meena et al. [18] incorporated the digital elevation model to improve landslide detection performance.

III. Methodology

A. Overview

As illustrated in Figure 1, HSSNet follows a symmetric U-shaped network architecture. Given the input remote sensing image, HSSNet initially utilizes ResNet as the encoder to extract spatial features. Subsequently, HSSNet designs a multiscale synergy attention (MSA) module to hierarchically collaborate multi-scale contextual features to identify landslide regions while suppressing interference from high-intensity backgrounds. The processed multi-scale features are then input into the decoder, where the spatial resolution is progressively restored, yielding the landslide detection results. To further optimize spatial details, HSSNet introduces the deformable spatial refining (DSR) module, which adaptively learns the irregular spatial structures of landslides. Through skip connections, these refined features are progressively fused with the encoder spatial features, allowing the spatial predictions to closely align with the true contours of the landslide.

B. Multiscale Synergy Attention

In landslide detection tasks, the excessive spatial down-sampling of CNNs results in inevitable damage to semantic information, making it hard to differentiate between landslides and high-intensity backgrounds. To alleviate this issue, MSA combines dilated convolutions with spatial-channel attention to enhance global-local features, facilitating landslide identification.

Dilated Convolution [19] enlarges the receptive field by inserting holes between the sampling points of the convolution kernel. Taking 2D spatial feature extraction process as an example, given an input feature x and a convolution kernel w, the dilated convolution operation in the feature map position i is formulated below:

$$y[p] = \sum_{k} w[k]y[i + r \cdot k]. \quad (1)$$

The y represents output feature map, k is the sampling positions of w. r determines dilation rate for sampling stride. Note that dilated convolution reduces to the standard convolution at $r = 1$. This operation enlarges the receptive field while keeping computational costs low, as the number of parameters stays same.

Considering the interference from high-intensity backgrounds, MSA designs spatial-channel attention mechanism to optimize multi-scale contextual features at different spatial resolutions, enhancing spatial representation and semantic discriminative ability.

$$y_{MSA} = conv(\{SCA(aconv_r(x))|r = 1,3,5,7\}), \quad (2)$$

where $aconv_r(\cdot)$ represents dilated convolution operation, $SCA(\cdot)$ denotes spatial-channel attention.

The spatial attention initially captures the key spatial representation by applying average and max pooling across the channel dimension, preserving the spatial resolution.

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These features are subsequently concatenated through a 1×1 convolution layer. The spatial attention features

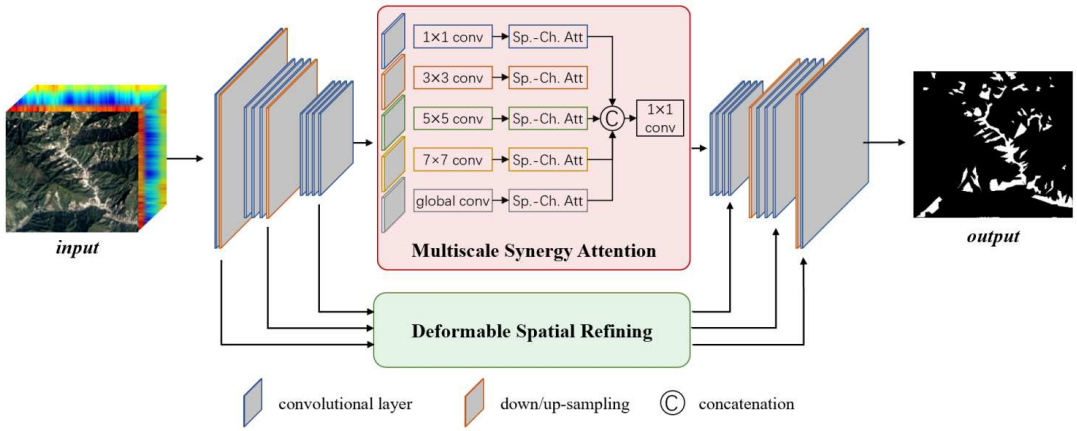


Figure 2. The framework of the proposed hierarchical spatial synergy network.

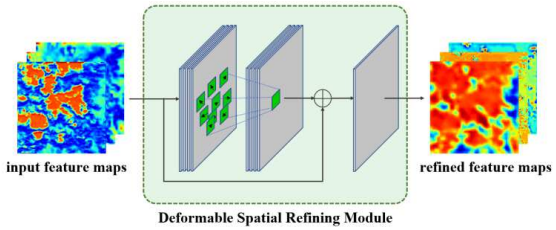


Figure 3. The details of the deformable spatial refining module.

y_{SA} are defined:

$$y_{SA} = x + conv([\mathcal{A}_{ch}(x), \mathcal{M}_{ch}(x)]), \quad (3)$$

where $\mathcal{A}_{ch}(\cdot)$ and $\mathcal{M}_{ch}(\cdot)$ represent average and max pooling in channel wise, respectively.

Then, HSSNet learns discriminative semantics by performing spatial wise average and max pooling, followed by a 1×1 convolution layer. After spatial pooling, y_{SA} is compressed into a 1D tensor. Channel attention combines these pooled features to represent semantic information at the image level. The channel attention features y_{CA} can be obtained:

$$y_{CA} = conv(\mathcal{A}_{sp}(y_{SA})) + conv(\mathcal{M}_{sp}(y_{SA})), \quad (4)$$

where $\mathcal{A}_{sp}(\cdot)$ and $\mathcal{M}_{sp}(\cdot)$ represent spatial wise average and max pooling, respectively.

Finally, these spatial and channel attention features are multiplied to form the optimized features:

$$y_{MSA} = conv(concat(y_{SA}, y_{CA})). \quad (5)$$

where $concat(\cdot)$ represents feature concatenation.

C. Deformable Spatial Refining

Landslide areas are typically influenced by the terrain, resulting in varying spatial dimensions and complex geometric structures, which pose challenges for precise spatial activation. However, since the fixed sampling positions, standard convolutions are inherently limited in modeling complex geometric transformations. Inspired by deformable convolution [20], Deformable Spatial Refining (DSR) is devised to represent spatial transformations, allowing the predicted results to closely align with the true landslide contours.

As shown in Figure 2, DSR utilizes residual structure with shortcut connections. The input of DSR is a set of encoded multi-scale features $\{x^{(l)}\}_{l=1}^{L}$, and the output is a corresponding set of refined spatial features $\{y^{(l)}\}_{l=1}^{L}$. The superscript (l) is layer index.

DSR consists of two deformable convolutional blocks to enhance spatial transformations. Each block contains four deformable convolutional layers. Unlike the fixed convolution sampling, DSR adjusts the sampling positions by learnable offsets $\{\Delta p_m \mid m = 1, \ldots, M\}$, where M is sampling positions count. The position p_0 on output feature map y is below:

$$y^{(i)}(p_0) = \sum_{p_n \in R} w^{(i)}(p_n) \cdot x^{(i)}(p_0 + p_n), \quad (6)$$

The p_n enumerates sampling positions in convolution grid R. In deformable convolution, equation 6 becomes:

$$y^{(i)}(p_0) = \sum_{p_n \in R} w^{(i)}(p_n) \cdot x^{(i)}(p_0 + p_n + \Delta p_m). \quad (7)$$

Sampling occurs at irregular positions $p_n + \Delta p_m$, where Δp_m represents the learned offsets obtained from additional convolutional layers, which capture the spatial structure of the landslide. This permits the sampling grid to deform irregularly, enabling complex structure modeling. The refined feature map is obtained:

$$y_{DSR}^{(l)} = conv(D(x^{(l)}) + x^{(l)}). \quad (8)$$

where $D(\cdot)$ represents the deformable convolution block, and $conv(\cdot)$ represents a 1×1convolution layer used for feature fusion. Finally, these refined spatial features are concatenated with decoder features via shortcut connection to progressively restore spatial details, resulting in the final landslide detection output.

IV. Experiments

A. Dataset

CAS [8] is a multi-sensor landslide dataset, consisting of 20,865 images from 9 regions. Rigorous quality assessment methods were employed in CAS to ensure its reliability and have been validated through experimental testing. In terms of data acquisition, public datasets

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provided by the Google Earth Engine platform and drone images from partners were primarily used. Pixel-level labeling was performed using QGIS and LabelMe software, with expert input and strict quality control. The image is provided in TIFF format with a 512×512 pixel resolution, comprising three subsets: images, labels, and masks. This paper follows standard procedures to randomly divide 16,692 and 4,173 images for training and testing.

B. Implementations

HSSNet uses the cross-entropy loss and structural similarity loss to optimize network training. HSSNet employs Stochastic Gradient Descent as the optimizer. The initial learning rate is 7×10^{-3} and weight decay coefficient is 1×10^{-4} to prevent overfitting. The learning rate adjustment strategy adopts cosine annealing mechanism.

HSSNet uses Accuracy, mean Intersection over Union (mIoU), F1-score, Precision, and Recall as quantitative evaluation metrics to assess model performance.

C. Ablation Studies

1） Impact of Different Backbone

Table 1 evaluates the different network architectures on detection performance. HSSNet achieves the best detection performance, using the deeper ResNet backbone network. MobileNet is designed for efficient operation that reduces computational load and model parameters to accelerate inference but limits its performance in handling complex visual tasks. Due to the demand for refined feature representation in landslide detection, MobileNet shows limited performance. Benefiting from the successive convolutional design, VGG obtains suboptimal detection performance in landslide detection tasks. ResNet introduces residual connections, stabilizing and enhancing the training of deep neural networks. HSSNet also incorporates skip connections to address the vanishing gradient problem, enabling efficient feature learning in very deep network structures. Therefore, HSSNet, using ResNet as the backbone, achieves optimal detection performance.

2） Effectiveness of the Proposed Modules

Table 2 validates the effectiveness of MSA and DSR. The MSA module focuses on the landslide regions at different scales and uses channel-space attention to automatically select more important contextual features for landslide identification. Compared to the baseline framework, MSA improves Precision and F1 by 5.20% and 5.55%, respectively. These results show that MSA, by synergizing multi-scale contextual features, enhances landslide identification and suppresses interference from high-intensity backgrounds. Further integrating DSR (+MSA+DSR), HSSNet achieves an obvious improvement. This indicates that DSR are useful to learn complex spatial structure.

3） Sensitive Analysis of Different Loss Functions

HSSNet using the BCE loss performed well in terms of accuracy, indicating that HSSNet can detect most landslide regions. However, mIoU equals to 79.11%, suggesting that there is a spatial mismatch when the model handles the segmentation of the background and landslide regions. Therefore, the BCE loss function primarily focuses on minimizing the difference between positive and negative samples, but it may not fully consider the structural and spatial information of the image. By considering the relative positions and structural relationships between pixels, BCE + SSIM better captures the edges and texture information of the landslide areas, thus improving detection accuracy. The notable gains in mIoU and recall rate indicate that SSIM contributes to better edge detection and the preservation of fine details.

D. Comparison Results

Table 4 compares HSSNet with various mainstream landslide detection models.

GCNet [21] uses vertical multi-convolutions and horizontal multipaths for training, which are reparametrized into a single convolution for inference. RDRNet [22] employs multi-path training architecture while reparametrizing them into single path for inference, boosting efficiency. However, GCNet and RDRNet emphasize network structural simplification, which is beneficial for speed and inference deployment, but may weaken the model ability to capture complex contextual relationships. DeepLabV3+ [19] expands the receptive field without compromising spatial resolution, excelling in mIoU and F1 score. As shown in Figure 3, while dilated convolutions expand the receptive field, DeepLabV3+ performs worse in small landslide segmenting, leading to missed detections. UNet uses skip connections to receive low-level spatial features, identifying more complete landslide regions, but it tends to produce false positives.

TABLE1 IMPACT OF DIFFERENT BACKBONE NETWORKS

Ablations	Accuracy	mIoU	F1	Precision	Recall
mobileNet	90.36%	76.37%	85.86%	86.36%	85.36%
VGG	93.94%	78.69%	88.25%	89.94%	86.63%
ResNet	96.68%	82.41%	90.10%	88.97%	91.26%

TABLE 2 EFFECTIVENESS OF THE PROPOSED MODULES

Ablations	Accuracy	mIoU	F1	Precision	Recall
baseline	90.34%	74.11%	81.84%	80.89%	82.82%
+MSA	95.39%	79.65%	87.39%	86.09%	88.75%
+MSA+DSR	96.68%	82.41%	90.10%	88.97%	91.26%

TABLE 3 SENSITIVE ANALYSIS OF DIFFERENT LOSS FUNCTIONS

Ablations	Accuracy	mIoU	F1	Precision	Recall
+BCE	92.36%	79.11%	88.87%	85.89%	90.36%
+BCE+SSIM	96.68%	82.41%	90.10%	88.97%	91.26%

TABLE 4 COMPARISON RESULTS

Methods	Accuracy	mIoU	F1	Precision	Recall
GCNet	80.93%	75.57%	84.52%	89.30%	80.93%
RDRNet	76.39%	71.84%	80.67%	87.13%	76.39%
DeepLabV3+	92.15%	79.74%	83.60%	82.86%	84.36%
UNet	94.31%	81.56%	85.98%	83.44%	85.51%
HSSNet	96.68%	82.41%	90.10%	88.97%	91.26%

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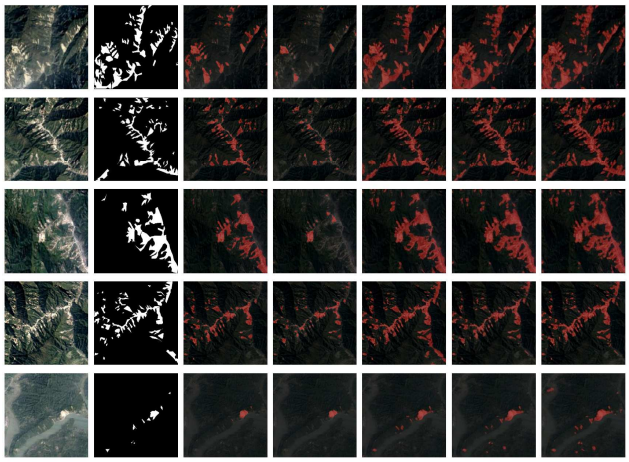


Figure 4. The comparison results of HSSNet.

By integrating the multiscale synergy attention and deformable spatial refining modules, HSSNet achieves the best performance. As shown in Figure 3, the spatial activation results closely matching true contours of landslide regions.

V. CONCLUSION

In satellite remote sensing imagery, landslide often exhibit highly irregular spatial patterns, and high-intensity background targets easily cause noise interference To address these issues, this paper proposes the Hierarchical Spatial Synergy Network (HSSNet). HSSNet designs a multiscale synergy attention module to optimize contextual features across multiple spatial scales, enhancing key landslide features while suppressing noise from high-intensity regions. Building upon this, HSSNet introduces a deformable spatial refining module to adaptively extract irregular features, improving the alignment between predicted and true contours. Experiments on the CAS public dataset demonstrate that HSSNet achieves performance comparable to mainstream detection methods.

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